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Automating Mammogram Reports Using Vision Language Models for Standardized, Clinically Actionable Insights

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***Abstract*—The newly increased mammography examinations and the shortage of expert radiologists are great problems encountered in real clinical settings of breast cancer screening. When screening is performed at a high volume, reporting inconsistencies, delayed reports, and the inability to derive structured information from free-text reports can have an impact on both the clinical workflow and downstream AI analysis. Clinical mammographic reports are mostly written in unstructured free text, making them less compatible with large-scale data analysis and automated AI systems. To address this, we propose a two-stage vision-language model pipeline which extracts five standardized features from mammographic images using constrained function calling, and then produces structured clinical PDF reports directly from the extracted features. The models we test are Gemini 2.5 Flash, Gemma 3 27B, Qwen3-VL 8B Instruct, and MedGemma 4B. Prompting alone does not result in consistent predictions across architecturally diverse models, as evidenced by the low inter-model Cohen's Kappa values for all feature pairs, not exceeding $\kappa = 0.23$. Nevertheless, Gemini 2.5 Flash outperforms the other models in the malignancy classification task, with an F1 score of 0.521, although MedGemma 4B, with domain-specific medical pretraining, has the lowest accuracy of the four models. The results validate the possibility of VLM-based mammographic reporting but highlight the importance of fine-tuning the VLM to the mammographic domain and providing radiological supervision prior to clinical use.**

***Index Terms*—Mammography, Breast Cancer Detection, Vision Language Models, Structured report generation, Feature Extraction, BI-RADS, Inter-model agreement, Cohen's Kappa, Clinical decision support, Medical Image Analysis**

I. Introduction

Breast cancer is one of the leading causes of cancer-related mortality worldwide, and early detection is a critical determinant of clinical outcome [1]. Screening mammography is considered the gold standard for detecting suspicious lesions, using descriptors such as mass shape, microcalcification patterns, architectural distortion, and breast density [3], [4]. These descriptors are formalized in the Breast Imaging Reporting and Data System (BI-RADS), which standardizes radiological terminology and conveys the risk of malignancy [5].

Despite the clinical importance of mammography, reporting is predominantly unstructured, resulting in inter-practitioner variability, limited interoperability with AI systems, and difficulty performing large-scale retrospective analyses [6], [7]. Unstructured reports hinder downstream applications such as automated triaging, decision support, and research-oriented analysis. Recent advances in large language models (LLMs) and vision-language models (VLMs) have enabled the integration of visual data with natural language, producing structured clinical outputs [8], [10]. Constrained function calling enforces strict output schemas, enhancing reliability and reducing hallucination [9], while medical pretraining of models, exemplified by MedGemma, aligns output with domain-specific clinical terminology [12]. Moreover, structured reporting not only facilitates AI-based decision support but also improves consistency in clinical practice, helping radiologists reduce inter-observer variability and standardize patient care. Additionally, structured outputs enable large-scale studies and benchmarking, which are critical for evaluating new diagnostic models and understanding population-level trends in breast cancer screening.

To evaluate this approach, we utilize an MLO-view subset of the RSNA Mammography dataset [16], providing high-quality annotated data for model assessment. The objectives of this study are as follows:

- To design a two-stage VLM-based automated report generation system for mammography.
- To use constrained function calling for the extraction of five standardized categorical mammographic features.

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- To generate structured clinical PDF and machine-readable CSV reports for downstream applications.
- To assess malignancy classification performance across Gemini 2.5 Flash, Gemma 3 27B, Qwen3-VL 8B Instruct, and MedGemma 4B models, including inter-model agreement and feature-level prediction stability.

II. Related Work

A. Deep Learning for Mammographic Image Analysis

The CNN-based methods have shown promising results in terms of lesion detection and malignancy classification, where they have learned discriminative features from pixel information directly, and in some cases, even equaled the performance of trained physicians on large well-curated data [14], [15]. An example of such model is the deep convolutional network created by Wu et al., which was trained on over 200,000 screening cases and achieved results on the level of expert radiologists in a multi-reader test [13]. However, such models are highly discriminative, as they provide classifications or bounding boxes but cannot create structured text for reporting purposes. Thus, the need to develop generative models that would be able to create text documents emerged due to this limitation.

B. Large Language Models in Clinical NLP

Transformers for large language models (LLMs) have greatly improved clinical natural language processing, allowing for entity extraction, generation of discharge summaries, and clinical text summarization using unstructured text [6], [7], [8]. In radiology, LLMs have been applied for conversion of free-text to structured fields, extraction of findings and impressions, and downstream use cases like coding and audit [7]. The key factor facilitating this has been function-calling or constrained generation, where the output is constrained to be within a defined schema through the API call itself, and reduces the chance of hallucination in contrast to free-text generation [9]. Clinical language models pretrained on domain-specific text have proven to be better aligned with clinical terms than general-purpose language models such as clinical BERT [9].

C. Vision-Language Models for Medical Imaging

Vision-Language Models (VLMs) represent both image and text data inputs, mapping medical images to structured findings and natural-language captions without hand-crafted features [10]. VLMs have been used for radiological report generation, pathology captioning and visual question answering tasks [10]. For mammography, models which have received medical pretraining, such as MedGemma, have proven to be clinically better aligned compared to general-purpose VLMs of the same scale [12]. AMRG represents a work most similar to our research, as it fine-tunes various VLM backbones with LoRA and generates mammography reports in an end-to-end manner, as well as comparing various backbones [17]. The main difference is that we test models in a zero-shot setting, separate feature extraction from the report generation into a two-step process, and calculate model agreements based on Cohen's Kappa and entropy.

III. Methodology

A. Dataset and Preprocessing

The dataset used for this experiment is "RSNA Screening Mammography Breast Cancer Detection" Kaggle competition held in 2022 [16], with corresponding metadata (`train.csv`), containing age, laterality, implants presence, BI-RADS breast density class(incomplete data) and binary cancer label (`cancer`: 0 or 1). Only Mediolateral Oblique (MLO) view images were chosen for analysis as it is the one giving the best view of breast tissues including axillary tail. The stratified subset of 100 images (50 malignant, 50 benign) is chosen for model comparison purposes. Conversion from DICOM to 8-bit PNG with normalization of pixel intensity in range $[0, 255]$ using `pydicom` and saving the image using PIL, followed by resizing of the image to maximum bounding dimensions of 1024×1024 pixels using Lanczos algorithm and preserving aspect ratio. Base64 encoding of the resized image for API based models and direct processing of PIL object for local MedGemma model using HuggingFace processor.

B. System Architecture and Pipeline Overview

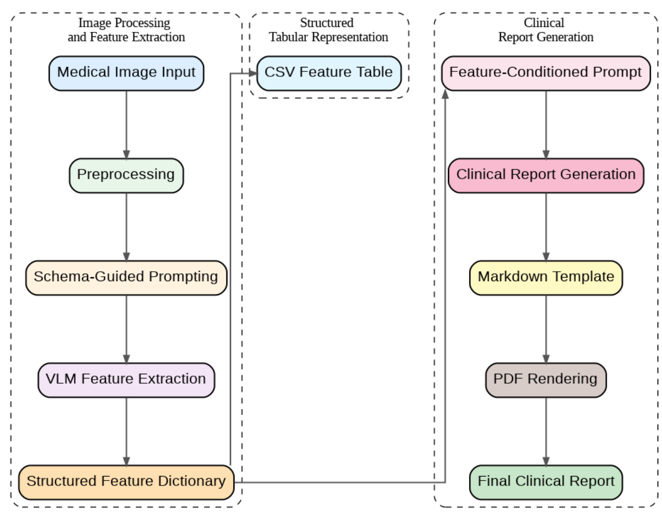


Fig. 1. Overview of the proposed pipeline organized into three modules: Image Processing and Feature Extraction (left), Structured Tabular Representation (centre), and Clinical Report Generation (right).

The proposed pipeline, illustrated in Fig. 1, is organised into three functional modules:

Image Processing and Feature Extraction:

- Images are resized to 1024×1024 pixels, normalized, and Base64-encoded for VLM input.
- A structured system prompt instructs the model to reason as a breast radiologist, prohibiting hallucination beyond observable features.
- Function-calling, leveraging schema-guided prompting constrains the output to five BI-RADS-aligned fields; models without native support use prompt-constrained generation with `ast` parsing.

- The extracted features are organized into a structured feature dictionary that serves as input to both CSV feature table and feature-conditioned report generation prompt.

Structured Tabular Representation:

- Per-image feature predictions are appended to a per-model CSV keyed by image ID.
- A log file enables resumable execution by skipping already-processed images.
- The uniform schema across all models supports direct cross-model comparison and downstream analytics.

Clinical Report Generation:

- Extracted features feed a feature-conditioned prompt for a second VLM call, generating a structured clinical narrative.
- The report covers six sections: Examination, Breast Composition, Findings, Impression, BI-RADS Assessment, and Recommendation.
- The output is rendered to PDF via XeLaTeX through `pypandoc`, with Markdown preserved as a fallback.

An orchestration layer manages rate limiting, error handling, and resumable execution across all three modules.

C. Model Selection and Access Methods

Four vision-language models were evaluated across the pipeline. Each model was accessed through a distinct interface reflecting differences in availability, deployment tier and native capabilities as summarized in Table I.

TABLE I
MODEL ACCESS METHODS AND INTERFACES

Model	Size	Interface	Endpoint / Backend
Gemini 2.5 Flash	–	Google GenAI SDK	Google AI API
Gemma 3	27B	OpenAI-compat. client	OpenRouter
Qwen3-VL	8B	OpenAI-compat. client	HF Router (Novita)
MedGemma	4B	Transformers (local)	On-device GPU

Gemini 2.5 Flash was accessed via the Google GenAI Python SDK using a Gemini API key. Extraction used the SDK's native function-calling interface, enforcing structured JSON output at the API level.

Gemma 3 27B was accessed via OpenRouter using an OpenAI-compatible client. Lacking native function-calling support here, extraction used *prompt-constrained generation*: the model was instructed to respond in a predefined function-call text format, parsed via regular expressions and `ast.literal_eval`.

Qwen3-VL 8B Instruct was accessed via the HuggingFace Inference Router (Novita AI backend) using an OpenAI-compatible client with a HuggingFace token. It natively supports function calling; `tool_choice` was forced to the extraction function for structured JSON output.

MedGemma 4B was run locally via HuggingFace `transformers` in `bfloat16` with automatic GPU/CPU placement. Lacking a function-calling API, it used the same prompt-constrained approach as Gemma 3.

All models used temperature 0.0 for deterministic, reproducible outputs.

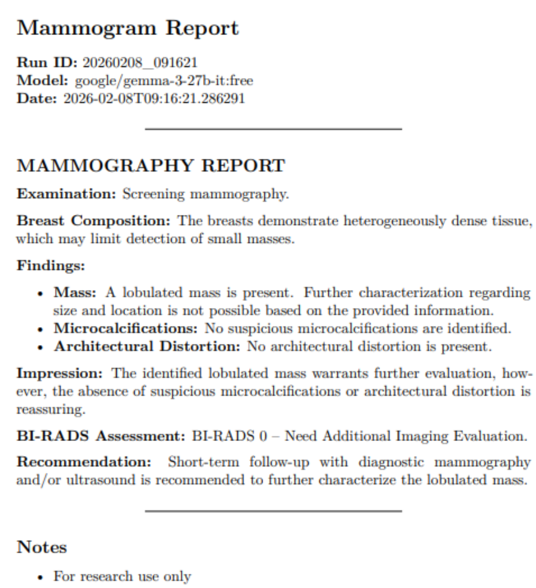
D. Structured Feature Extraction

The first stage of the pipeline frames mammographic feature extraction as a constrained structured-output task. A shared feature schema defines five categorical output fields drawn from standard BI-RADS terminology [5]:

- **Mass shape:** `round` | `oval` | `lobulated` | `irregular` | `stellate`
- **Microcalcification pattern:** `none` | `scattered` | `clustered` | `linear` | `segmental` | `pleomorphic`
- **Architectural distortion:** `none` | `mild` | `moderate` | `severe`
- **Breast density category:** `fatty` | `scattered` | `heterogeneously dense` | `extremely dense`
- **Malignancy label:** `yes` | `no`

For models that support function calling natively (Gemini 2.5 Flash, Qwen3-VL), the schema was defined as a JSON tool definition, and to ensure the output is valid structured JSON, `tool_choice` was passed to the extraction function. For models that do not have support for this (Gemma 3, MedGemma), the same schema was placed in a system prompt that required a function-call-style text response, which was then matched with `re` and parsed into a structured dictionary using `ast.literal_eval`. In all cases, the system prompt tells the model to play the role of a board-certified breast radiologist, to only use image features that it can see, to not hallucinate, and to represent features it could not see with a categorical null value (e.g., `"none"`).

E. Structured Report Generation


Mammogram Report

Run ID: 20260208_091621
Model: google/gemma-3-27b-it:free
Date: 2026-02-08T09:16:21.286291

MAMMOGRAPHY REPORT

Examination: Screening mammography.

Breast Composition: The breasts demonstrate heterogeneously dense tissue, which may limit detection of small masses.

Findings:

- **Mass:** A lobulated mass is present. Further characterization regarding size and location is not possible based on the provided information.
- **Microcalcifications:** No suspicious microcalcifications are identified.
- **Architectural Distortion:** No architectural distortion is present.

Impression: The identified lobulated mass warrants further evaluation, however, the absence of suspicious microcalcifications or architectural distortion is reassuring.

BI-RADS Assessment: BI-RADS 0 – Need Additional Imaging Evaluation.

Recommendation: Short-term follow-up with diagnostic mammography and/or ultrasound is recommended to further characterize the lobulated mass.

Notes

- For research use only

Fig. 2. Example mammography report generated from structured feature inputs using the proposed VLM-based pipeline.

Fig. 2 presents a sample mammography report generated by the proposed pipeline. The second stage takes the extracted feature dictionary and generates a complete, dictation-style clinical mammography report. The feature JSON is embedded directly into a system-level report prompt that instructs the model to produce a report conforming to the following required sections: Examination, Breast Composition (translated into BI-RADS density categories), Findings (mass morphology, calcification pattern, architectural distortion), Impression, BI-RADS Assessment Category and Recommendation. The model is explicitly forbidden to offer findings not found in the input features, speculate laterality or lesion size, or use non-clinical terms. With the XeLaTeX engine, `latex2pdf` is used to convert laTeX and PDF. When there is an error in the rendering of the PDF, a Markdown source is saved as an artifact in case the formatted version is lost.

F. Batch Evaluation Protocol

Images were processed using a recursive scan of all PNG files. Resumable processing was achieved through a simple log file, which records the IDs of successfully processed images; on restart, previously processed images are skipped. An inter-request delay (default 2s) was implemented between calls to respect API rate limits. A feature table was obtained by appending each image's per-model prediction to a CSV (e.g., `mammogram_features_gemini.csv`), keyed by image stem ID.

IV. Results and Discussion

A. Malignancy Classification Performance

TABLE II
MALIGNANCY PREDICTION PERFORMANCE ACROSS MODELS

Model	**Acc.**	**Prec.**	**Rec.**	**F1**	**n**
Gemini 2.5 Flash	**0.540**	0.543	**0.500**	**0.521**	100
Gemma 3 27B	0.515	**0.545**	0.122	0.200	99
Qwen3-VL 8B Instruct	0.515	0.538	0.143	0.226	99
MedGemma 4B	0.474	0.489	0.440	0.463	97

Table II shows the binary classification performance metrics for the malignant field, which is the only labeled feature considered here. Out of the 100 images, $n = 100, 99, 99$, and 97 have been processed by Gemini, Gemma, Qwen, and MedGemma, respectively, with the rest being excluded because of failed structured output parsing (see Section III-F). With the equal 50/50 split of the evaluation sample prevalence, any classifier will achieve 50% accuracy if it predicts "benign" in every case, whereas Gemma (0.515) and Qwen (0.515) surpass this benchmark marginally, MedGemma (0.474) undershoots it, and Gemini (0.540) exceeds chance-level accuracy notably.

As one can see from the results, the approaches to prediction of each model differ significantly. Gemini 2.5 Flash demonstrates the best accuracy (0.540), recall (0.500), and F1-score (0.521) among the four models, with its precision (0.543) nearly matching the precision of Gemma. Gemma 3 27B and Qwen3-VL 8B Instruct generate similar scores in terms of accuracy (0.515) with rather low recall (0.122 and 0.143, respectively) that means their conservative low-sensitivity approach. MedGemma 4B provides the lowest accuracy (0.474) among the four models, yet has fairly well-balanced precision and recall (0.489 and 0.440, respectively). Such trade-offs are illustrated in the following figure 3 on all four metrics.

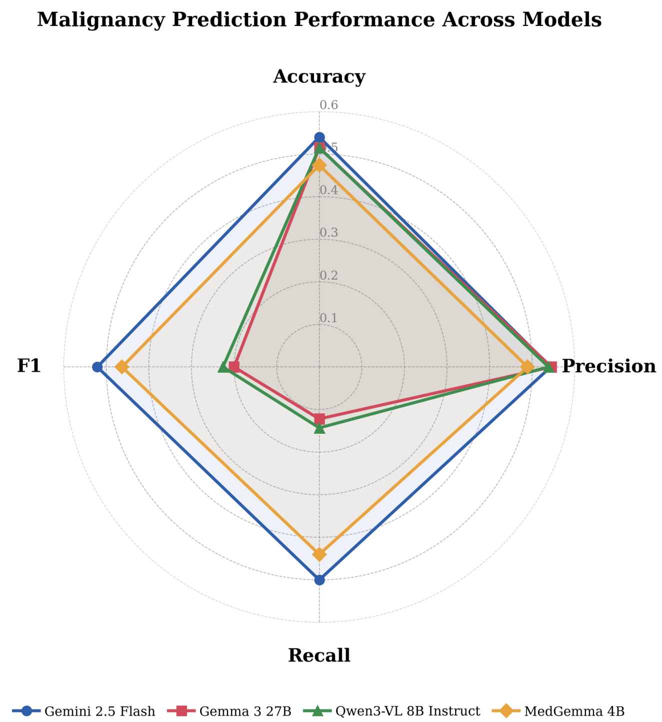


Fig. 3. A radar diagram of accuracy, precision, recall and F1 score over the four models on the task of malignancy prediction.

B. Inter-Model Agreement

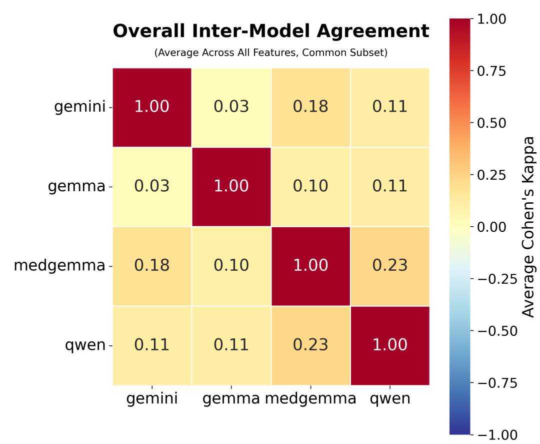


Fig. 4. Pairwise inter-model Cohen's Kappa heatmap averaged across all five features. Values close to zero indicate near-chance agreement between model pairs.

Cohen's Kappa scores for pairwise agreement among all five extracted features were calculated for the data subset common to all four models, as shown in Fig. 4. Off-diagonal elements are consistently low, lying within 0.03–0.23, showing that there is no substantial agreement between models even though they are using the same extraction protocol. The maximum agreement can be seen between MedGemma and Qwen ($\kappa = 0.23$) and the minimum agreement between Gemini and Gemma ($\kappa = 0.03$). It can be concluded that the models use mammogram features differently. One needs to keep in mind that the Kappa score measures agreement between models and not between models and ground truth. Perfectly agreeing models could both be clinically wrong.

C. Feature-Level Stability Analysis

Each feature is plotted in a two-dimensional phase space in Fig. 5, using mean inter-model agreement and mean prediction entropy. `malignant` shows the highest agreement (0.80) and lowest entropy (0.52). `Architectural_distortion` and `microcalcification_pattern` cluster together with moderate agreement and higher entropy. `mass_shape` shows the lowest agreement (0.50) and highest entropy (1.40), the least stable region of the diagram, while `breast_density_category` falls in between. This comparison is not on equal footing, however: `malignant` is binary, while `mass_shape` has five categories, so lower chance-level agreement and higher entropy for `mass_shape` are expected by construction, independent of genuine interpretive difficulty. Its apparent instability should therefore be read as an upper bound on ambiguity rather than a cardinality-adjusted measure of it. With this caveat, the results still suggest ambiguous images and higher-cardinality features are harder for VLMs to classify consistently without radiologist supervision.

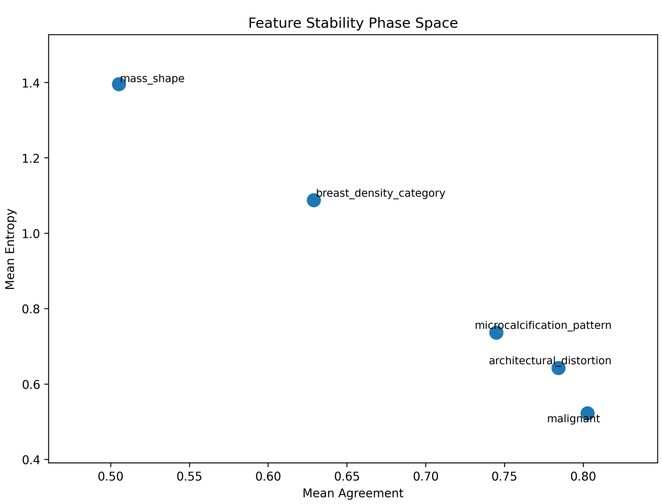


Fig. 5. Feature stability phase space mapping mean inter-model agreement against mean entropy for each extracted feature. Features in the lower-right quadrant are most stable; upper-left indicates highest ambiguity.

D. Image-Level Disagreement and Entropy

Fig. 6 and Fig. 7 show the distributions of per-image disagreement and entropy scores across the evaluation set. The disagreement distribution peaks around 0.55–0.60 ($n = 13$), with a mean of 0.481 ($\sigma = 0.134$) and scores ranging from

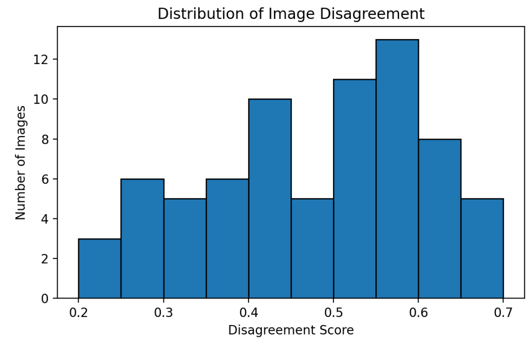


Fig. 6. Distribution of image-level disagreement scores across the evaluation set.

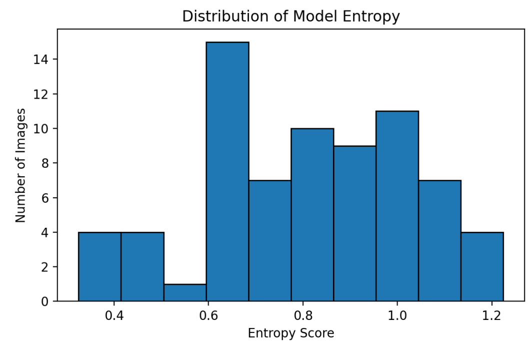


Fig. 7. Distribution of per-image model entropy scores.

0.200 to 0.700. No image in the evaluation set achieves a disagreement score ≤ 0.15, indicating that cross-model divergence is pervasive rather than confined to a few difficult cases. The entropy distribution is wider and roughly bimodal, with peaks near 0.60–0.65 and 1.00, a mean of 0.806 ($\sigma = 0.223$), and a range of 0.325–1.225. Together, these distributions confirm that most images present genuine interpretive ambiguity for current VLM-based pipelines. The most controversial images (disagreement score $= 0.70$) represent cases where the four models agreed on almost none of the extracted features, underscoring the challenge of zero-shot mammographic interpretation.

E. t-SNE Visualisation of Interpretation Space

Fig. 8 and Fig. 9 show the t-SNE projections of the combined per-image feature predictions from the four evaluated VLMs, coloured by prediction entropy and model disagreement, respectively. Each point represents a mammogram image, and spatial proximity indicates similarity in the multi-model feature prediction patterns. Both projections reveal three major clusters, suggesting that the models collectively group mammogram images into interpretively distinct regions despite limited feature-level agreement. The upper-left cluster contains several high-entropy and high-disagreement samples, indicating more ambiguous cases with unstable model interpretations, while the compact lower-right cluster shows comparatively lower uncertainty and greater prediction consistency. Overall, the t-SNE analysis shows that image-level uncertainty is not randomly distributed but partially reflected in the latent struc-

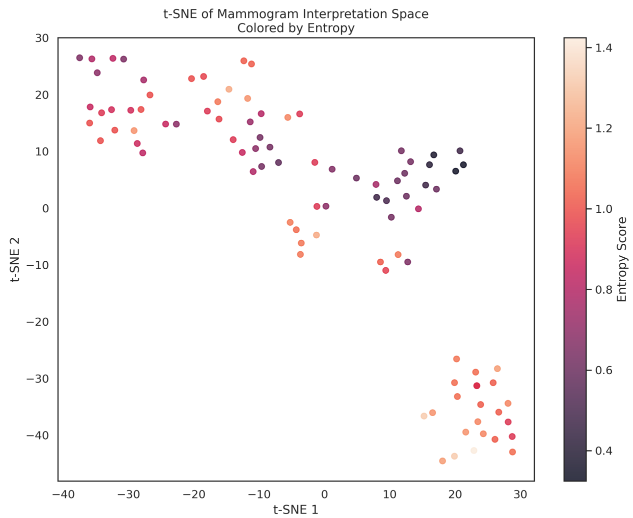


Fig. 8. t-SNE projection of mammogram interpretation space coloured by per-image entropy score. High-entropy (uncertain) images cluster in the upper-left region.

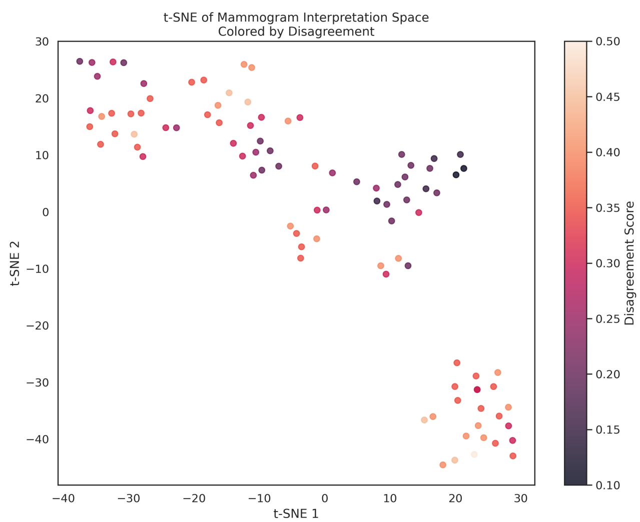


Fig. 9. t-SNE projection of mammogram interpretation space coloured by inter-model disagreement score.

ture of the multi-model prediction space. This supports the use of embedding-based visualization for identifying difficult mammographic cases and assessing the reliability of VLM-driven reporting.

V. Conclusion

This paper introduces a structured VLM-based pipeline to automate mammographic interpretation and report generation that gives machine-readable outputs with CSV and PDF via constrained function calling. The best performing F1 score (0.521) was obtained by Gemini 2.5 followed by MedGemma 4B (0.463) whereas the agreement between the different models on the common feature set was low, with Cohen's Kappa ranging from 0.03 to 0.23. The results indicate that some models were more successful in predicting the malignancy label, but the predictions were not necessarily consistent across models. Our framework shows the promise of VLM-assisted mammographic reporting, but the ability to associate density labels and generate reports with ground truth or radiologist review was not evaluated, and the evaluation employed a balanced set of 100 images instead of a prevalence of real-world cases, meaning the results are not indicative of clinical readiness. This highlights the importance of fine-tuning the AI model to the specific domain and prospective validation by radiologists.

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